Community Detection in Bacterial and Viral Networks

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Abstract

This paper investigates community structure in bacterial and viral protein similarity networks using modularity-based community detection methods. We compare several algorithms and perform a sensitivity analysis on the resolution parameter using Adjusted Rand Index and Adjusted Mutual Information.

1 Introduction

1.1 Motivation

1.1.1 Why statistical network analysis is important

We live in a highly interconnected world where many physical, social, technological and biological systems consist of agents or entities interacting with each other. Examples include a virus being transmitted over social contact networks, global trade between countries, and the human brain. Any such system can be represented as a network by denoting the entities as nodes and the interactions between them as edges. This makes networks an important type of data spanning a remarkable variety of complex systems. It is therefore very important to have mathematically rigorous and practically useful methods for statistical analysis of networks.

1.1.2 Biological Networks

Biological systems are often represented as networks. Essentially, every biological entity has interactions with oder biological entitites, from th emolecular to the ecosystem level, providing us with the oppurtunity to model biology using many different types of netowrks such as ecological, neurogical, metabolic and protein-protein networks.

A protein-protein interaction network (PPI network), also known as the interactome, is a type of biological network where a node represents a protein and an edge represents the physical interaction between each pair of proteins.

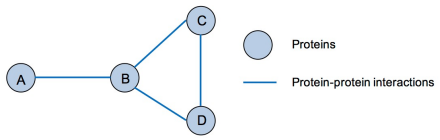

Figure 1: Protein–protein interaction network. Adapted from European Bioinformatics Institute (EMBL-EBI) 2024.

1.1.3 Why Community Detection

M. E. J. Newman (2006) argues that detecting and characterizing community structure is one of the fundamental problems in the study of networked systems, and notes that many networks of scientific interest — including biological ones — naturally divide into communities. Detecting these communitys can help us understand the structure and function of the network, as well as the underlying system it represents. For example, in the context of PPI(protein-protein interaction) networks, communities can help identify unknown protein complexes.

1.2 Definition of a Network

1.2.1 Graph as a mathematical object

We call a collection of connected objects a network. In the field of mathematics, a network can also be called a graph. The mathematical theory behind networks is known as graph theory. A graph is a mathematical object that consists of a set of objects (called vetices or nodes) that are connected by links (called edges).

1.2.2 Adjacency list

An adjacency list is a graph representation method that uses an array of lists to store connections between verticec. Each array index represents a vertex, and its corresponding list contains all neighboring nodes, making it highly efficient for sparse graphs($G = (V, E)$, where the number of edges is much less than the maximum possible number of edges). The main operation performed by the adjacency list data structure is to report a list of neighboprs of a given vertex.

1.2.3 Adjacency matrix

The main alternative to the adjacency list is the adjacency matrix, a matrix whose rows and columns are indexed by vertices and the cell (i, j) contains 1 if there is an edge between vertices i and j, or 0 if no such edge exists. The diagonal elements of the matrix are all 0, since edges from a vertex to itself are not allowed in simple graphs.

1.2.4 Directed vs. undirected networks

Undirected and directed graohs are fundamental concepts in grap theory. An undirceted graph is a graph where the edges do not have a specific direction and it is bidirectional in nature meaning it does not have a parent-child relation concept. In contrast, a directed graph is a graph where the edges have a specific direction, indicating a one-way relationship between the vertices.

An important difference between directed and undirected graphs is that in a undirected graph, relationships are symmetric, meaning if vertex A is connected to vertex B, then vertex B is also connected to vertex A. In a directed graph, relationships are asymmetric, meaning if vertex A is connected to vertex B, it does not necessarily imply that vertex B is connected to vertex A. This Characteristic is well represented in the adjacency matrix as you can see in the figure below.

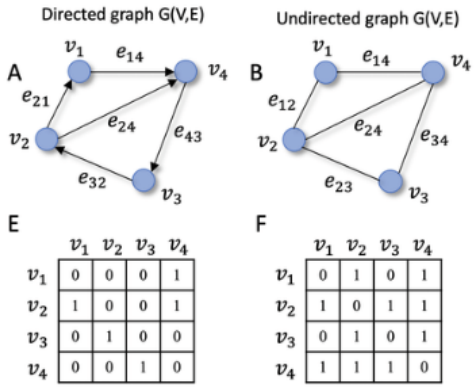

Figure 2: Directed vs. undirected graph. Adapted from ResearchGate 2020.

1.2.5 Weighted undirected network

Because in this paper we will be working with weighted undirected networks, we will now define what a weighted undirected network is. A weighted undirected network is a graph where each edge has a weight associated with it, representing the strength or importance of the connection between two vertices. In such networks, the adjacency matrix is not binary but contains numerical weights for each edge. Mathematically, a weighted undirected network can be represented as $G = (V, E, W)$, where V is the set of vertices, E is the set of edges, and W is a function that assigns a weight to each edge in E.

1.3 Network Analysis

1.3.1 Descriptive statistics

Hevey (2018) states that Network analysis provides the capacity to estimate complex patterns of relationships and the network structure can be analysed to reveal core features the network. A variety of basic descriptive metrics can be used to characterize network topology, including number of nodes and edges, network density, degree distribution and centrality measures. These measures provide an initial understanding of connectivity patterns and structural organization.

1.3.2 Community Detection and Community Comparison

In network science, a community is a group of nodes that are more densely connected to each other than to the rest of the network. Detecting communities can help us understand the functional organization of a system. For example, in a PPI network, groups of proteins that interact often perform the same function in a cell. Community detection methods such as Modularity, Louvain and Leiden algorithms can be used to identify these community structures. For this paper, attention is given to similarity measures for community comparison such as Adjusted Rand Index (ARI) and Adjusted Mutual Information (AMI), which quantify the agreement between different community structures. All these concepts will be explained in more detail in the following sections.

2 Network Summary Statistics

2.1 Basic Metrics

2.1.1 Number of nodes and edges

The number of nodes and edges are fundamental metrics that provide a basic understanding of the size and complexity of a network. The number of nodes, denoted as $|V|$, represents the total count of distinct entities in the network, while the number of edges, denoted as $|E|$, represents the total count of connections between those entities.

2.1.2 Density

Network density is a measure of how many edges are in the network compared to the maximum possible number of edges. THe metric is defined as the ratio of observed edges to the total possible connections, ranging from 0 to 1, where 0 indicates a completely disconnected network and 1 indicates a fully connected network. For an undirected network, the density can be calculated using the formula:

$$\text{Density} = \frac{|E|}{\binom{|V|}{2}} = \frac{2|E|}{|V|(|V|-1)}$$

2.1.3 Average degree

The average degree of a netork is a network metric representing the mean number of connections per node. The degree of a node refers to the number of edges connected to that node. These edges are called incident edges, meaning edges that are directly connected to the node in question. In an undirected network, the degree simply counts all incident edges, while in a directed network, a distinction is made between in-degree and out-degree. For an undirected network, it can be expressed mathematically as:

$$\text{Average Degree} = \frac{2|E|}{|V|}$$

For an directed network, it can be expressed mathematically as:

$$\text{Average In-Degree} = \text{Average Out-Degree} = \frac{|E|}{|V|}$$

Average In-Degree means the average number of incoming edges per node, while Average Out-Degree means the average number of outgoing edges per node. They are equal in a directed network because each edge contributes to both an in-degree and an out-degree.

2.2 Degree Distribution

2.2.1 Definition of degree distribution

The degree distribution of a networ describes the probability distribution of the degrees of nodes in a network. It specifies the probability P(k) that a randomy selected node has exactly k connections (or degree k). The distribution provides insighjt into the structural organization of a network and allows the differentiation between homogeneous and heterogeneous networks.

2.2.2 Random graph and scale-free graph

Random graphs and scale-free graphs are two fundamental network models that differs significantly in their degree distributions. A random graph, often modeled using the Erdős–Rényi framework, is construced by connecting nodes randomly with a fixed probability p. In such networks, edges are formed independently of each other. As a result, most nodes tend to have a

similar number of connections, leading to a relatively homogeneous structure. Highly connected nodes (hubs) are extremely rare in random graphs. The degree distribution of a random graph follows a Poisson distribution, which implies that most nodes have a degree close to the average degree. The probability function for a Poisson distribution is given by:

$$P(k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

where λ is the average degree of the network.

Scale free graphs, on the other hand, exhibit a highly heterogeneous structure. These networks are grow over time, and new nodes preferentially attach to alread welll-connected nodes. This process leads to the emergence of hubs, which are nodes with a significantly higher degree than the average node. Consequently, scale-free networks are characterized by strong degree ineuqality. Scale-free networks follow a power-law degree distribution of the form:

$$P(k) \sim k^{-\gamma}$$

where γ is a positive constant typically in the range of 2 to 3. This power-law behavior indicates that while most nodes have a low degree, a few nodes (hubs) have a very high degree. Below is a visual representation of the degree distribution for random and scale-free networks.

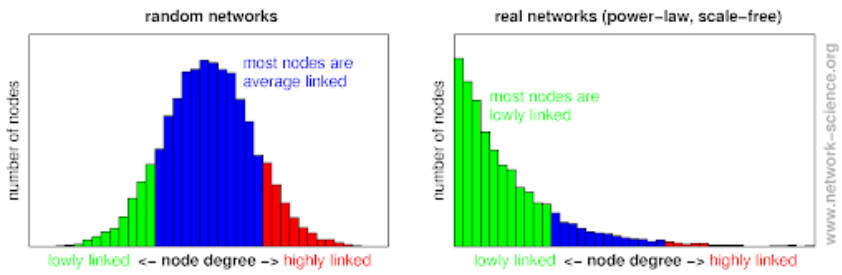

Figure 3: Comparison of random and scale-free networks and their corresponding degree distributions. Adapted from Science n.d.

2.2.3 Structural Implications

The degree distribution of a network has significant implications for its structure and dynamics. In networks with a Poisson degree distribution, connectivity is relatively uniform, meaning most of the nodes have the the same amount of connections and there are rarely outliers. Therefore, the network does not depend heavily on individual nodes. In contrast, scale-free networks are dominated by hubs, which play a crucial role in maintaining overall connectivity. The structural difference affects various network properties. For instance scale-free networks are typically robust against random node failures, as most nodes are low-degree nodes. However, they are highly vulnerable to targeted attacks on hubs. Furthemore, the presence of hubs strongly influences centrality measures and also modularity. As a result, understanding the degree distribution is essential for intepreting community detection results and structural properties. Centrality measures and Modularity will be discussed in the following sections.

2.3 Centrality Measures

2.3.1 Degree centrality

Degree centrality measures a node's importance in a network by counting its total number of connections (degree). In an undirected network, the degree centrality of a node is simply the number of edges connected to it. It can be expressed mathematically as:

$$C_D(v) = \deg(v)$$

where $\deg(v)$ is the degree of node v. In a directed network, degree centrality can be further divided into in-degree centrality and out-degree centrality. It can be expressed mathematically as:

$$C_{D_{in}}(v) = \deg_{in}(v)$$
$$C_{D_{out}}(v) = \deg_{out}(v)$$

where $\deg_{in}(v)$ is the number of incoming edges to node v and $\deg_{out}(v)$ is the number of outgoing edges from node v.

2.3.2 Betweenness centrality

Betweenness centrality measures the importance of a node by quantifying how often it lies on shortest paths between other pairs of nodes. More formally, it counts the proportion of shortest paths between all node pairs that pass through a given node. Mathematically, the betweenness centrality of a node v is defined as:

$$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

where σ_{st} is the total number of shortest paths from node s to node t, and $\sigma_{st}(v)$ is the number of those paths that pass through node v. Nodes with a high betweeness centrality often act as bridges between different parts of the network, frequently connecting otherwise weakly connected communities. Ednge betweenness extends the same idea to edges instead of nodes. It measures how often an edge lies on the shortest paths between pairs of nodes. The betweenness centrality of an edge e is defined as:

$$C_B(e) = \sum_{s \neq t} \frac{\sigma_{st}(e)}{\sigma_{st}}$$

where σ_{st} is the total number of shortest paths from node s to node t, and $\sigma_{st}(e)$ is the number of those paths that pass through edge e. This characteristic is particularly important in community detection methods based on edge betweenness, such as the algorithm introduced by M. G. Newman M. E. J. (2002), which will be discussed in more detail in the section on community detection methods.

2.4 Modularity

The core idea behind modularity is, a network has a strong community structure if:

- there are many edges between nodes within the same community.
- there are few edges between nodes in different communities.

Therefore, modularity compares the observed structure to a random baseline. This means that modularity measures how much more densely connected nodes within the same community are than we would expect by chance. The modularity formula is given by:

$$Q = \frac{1}{2m} \sum_{ij} (A_{ij} - P_{ij}) \delta(c_i, c_j)$$

where m is the total number of edges in the network, A_{ij} is the adjacency matrix element representing the presence of an edge between nodes i and j, P_{ij} is the expected number of edges between nodes i and j in a random graph with the same degree distribution, and $\delta(c_i, c_j)$ is the Kronecker delta function that equals 1 if nodes i and j are in the same community (i.e., $c_i = c_j$) and 0 otherwise. The modularity value ranges theoretically between -1 and 1, although in practice it typically lies between 0 and 1. Higher modularity values indicate a stronger community structure, meaning that more edges fall within communities than would be expected in a comparable random network. A known limitation of modularity is the resolution limit problem, meaning that small communities may not be detected because they do not contribute enough to the overall modularity score.

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